Geophys. J. R. astr. Soc. (1974) **37**, 245–262.

Remanent Magnetization of a 50 m Core from the Moenkopi Formation, Western Colorado*

Czang-go Baag and C. E. Helsley

(Received 1973 November 12) †

Summary

The NRM of 1117 specimens cut from a core taken from a 49·7 m inclined bore in the Lower Triassic Moenkopi Formation revealed one normal and two reversed magnetic zones. Samples were oriented using the ellipticity of the bedding plane sections cut by the core. The NRM directions of samples from the core are better clustered than those previously measured in surface samples from the same stratigraphic interval. This suggests that much of the scatter observed in the surface samples was due to surface effects, such as weathering.

The estimate of the concentration parameter, K, increases with decreasing grain size. Storage tests showed that specimens with low intensities (lower than $0{\cdot}2 \times 10^{-5}$ emu cm^{-3}) are generally the most unstable. This is considered to be due to larger viscous moments in these specimens, possibly associated with superparamagnetic components. Thermal demagnetization resulted in a change in the mean field direction for each polarity zone of less than 5° accompanied by a slight decrease in the dispersion.

Introduction

Often a serious problem in the study of palaeomagnetism is the scatter of natural remanent magnetizations (NRM) caused by either stable or unstable secondary magnetizations. These can apparently be related to the geological history of the rocks or their mode of surface occurrence. Throughout an entire rock formation, rocks can acquire a stable ancient secondary magnetization by internal processes (e.g. Briden 1965, 1967; Chamalaun 1964; Farrell & May 1969; Ku *et al.* 1967; Irving & Opdyke 1965; Storetvedt 1968). Rocks on, or close to, the surface can acquire secondary magnetizations due to surface effects (Irving 1964), whereas samples recovered from a road cut do not (e.g. Graham & Hales 1961). In this article surface samples refer to rock samples collected from natural exposures and not from recent deep road cuts.

Palaeomagnetists want to avoid the problems of surface effects on samples, both in the field and in the laboratory. It is often desirable to collect reliable rock samples from regions of poor and/or insufficient outcrops. These problems are obviously avoided when collecting subsurface samples.

A long core taken from a 50 m inclined bore in the Lower Triassic Moenkopi

* Contribution No. 248, Institute for Geological Sciences, The University of Texas at Dallas.

† Received in original form 1973 September 6.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

Formation was oriented using the ellipticity of the core's bedding plane, which is believed to be similar to the method referred to by Briden & Ward (1966). In this article we will show that the orientation error introduced by this method is not significant. We will also show that the NRM of the subsurface samples contains very little secondary component.

Sampling techniques

The drill site was located approximately 16 km south of Gateway, Colorado (Fig. 1). This site was selected because the surface outcrops had been extensively studied (Helsley 1969) and within the horizontal and vertical range of the drill the Moenkopi Formation lacked large scale cross-bedding, significant thicknesses of massive bedding, or slumping structures which could have obscured the bedding plane. Moreover, at this site the dip of the Moenkopi Formation is almost horizontal.

A 50 m inclined hole (Fig. 2) was drilled using a portable diamond drill (Winkie, Model GW 15) from which a 2·5-cm diameter core was recovered. The core was drilled at an angle of 58° in the direction N 30° W. Core segments were recovered at 1–3 m drilling increments and their up directions were immediately marked on their sides. Even with great care, there was still the possibility that short segments could have been inverted when they were removed from the core barrel. Adjacent pieces were glued together across interlocking broken surfaces whenever possible. Each core segment was then oriented in the following manner.

(1) Points were marked on the upper surface of the segment, at the extremity of each ellipse formed by the intersection of the bedding planes and the core surface (see Fig. 3).

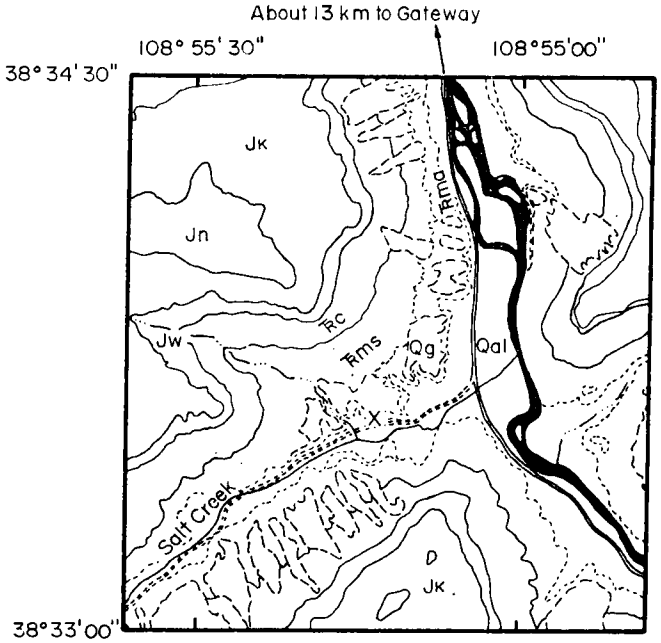


FIG. 1. Location map showing the drilling site. X = the drilling site; TRma = Ali Baba member; TRms = Sewemup member of the Moenkopi Formation; TRc = Chinle Formation; Jw = Jurassic Wingate sandstone; Jk = Jurassic Kayenta Formation; Jn = Jurassic Navajo sandstone; Qg = Terrace gravel; Qal = Alluvium.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

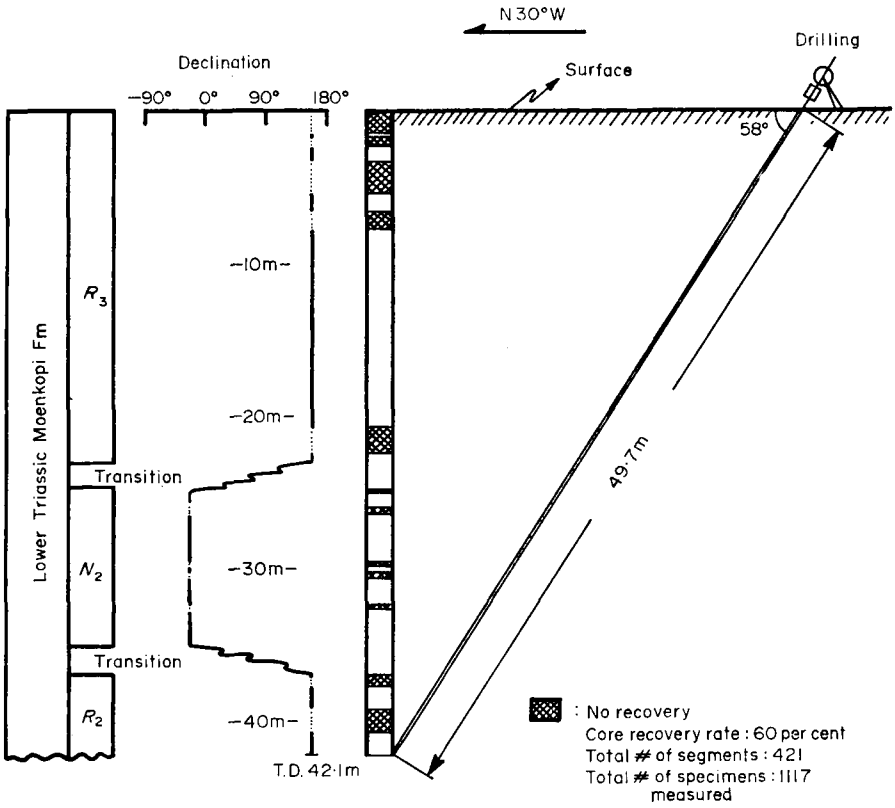


FIG. 2. Schematic diagram showing the cross-section of the drilling hole and its magnetic polarity.

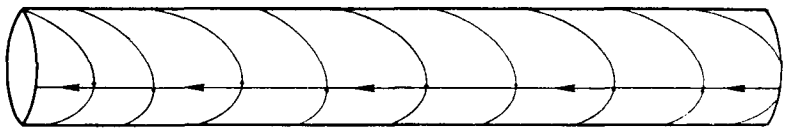


FIG. 3. Schematic diagram illustrating the orientation of a core segment. Curved lines represent ellipses formed by the intersection of the bedding plane and the core surface; the points are marked at the extremities of the ellipses; the straight line represents the orientation of the core segment; and the arrows represent the up direction of the core segment.

(2) A line representing a 'best fit' was drawn along the core joining these points and was taken as the orientation mark of the core (relative to the bedding plane), corresponding to the line made during the normal short core orientation procedure.

(3) The segments were cut at 2·4-cm intervals into specimens which were subjected to palaeomagnetic analysis.

Descriptions of core samples

The geology at the drilling site and vicinity has been described by Shoemaker (1955), Shoemaker & Newman (1959), and Helsley (1969). The hole bottomed at 42·13 m in the middle of Ali Baba member of the Moenkopi Formation.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

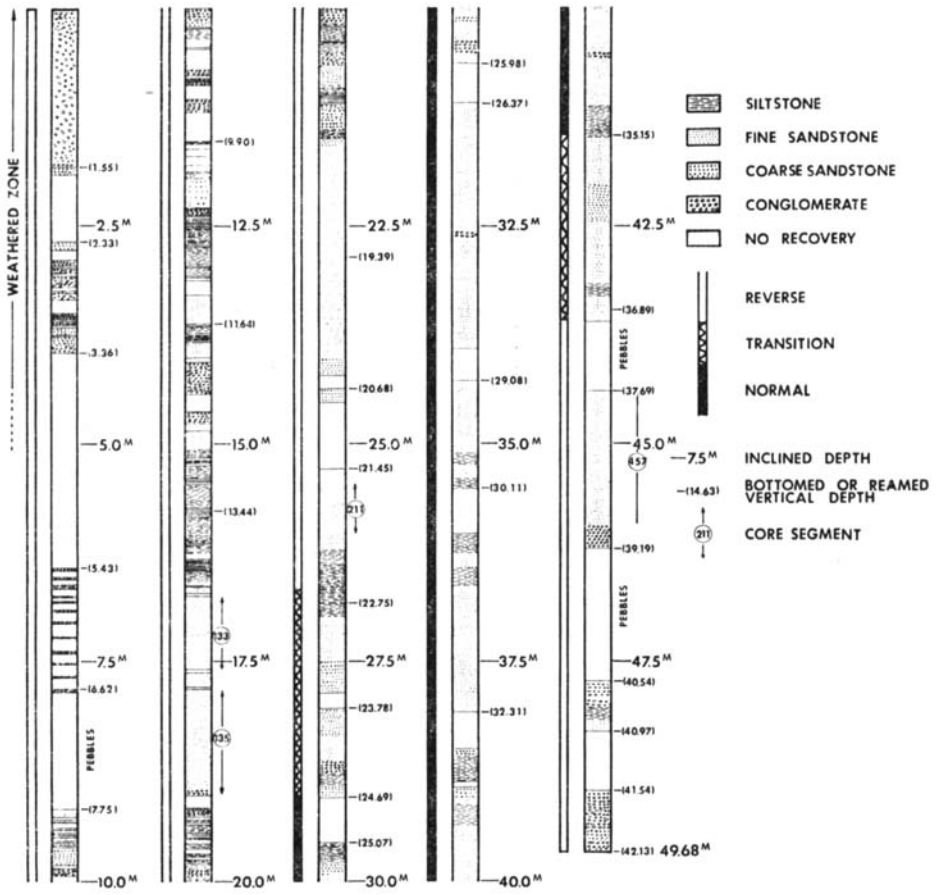


FIG. 4. Columnar section of core samples showing lithologic types and intervals where no core was recovered.

Fig. 4 illustrates the stratigraphy of the cored interval. The recovered core included about 60 per cent of the rock column. The missing sections have been associated with the conglomeratic units encountered in the coring.

Core was recovered after drilling increments of 1–1·5 m, in the upper half of the hole, and increments of 2–3 m in the lower half. For the upper half of the core, the relative depth position of the individual segments within each increment was approximated by inspecting the surface at the end of each segment. For the lower half of the core it was not possible to locate the depth position of many of the segments. It was therefore assumed, as a matter of convention, that all losses were confined to the lower portion of the respective increment.

From the 49·7 m inclined hole, the core was recovered in 421 independently oriented segments and cut into 1117 specimens. These were studied for their magnetic properties using a PAR SM–1 magnetometer.

Stability test

(a) *Storage test*

The NRM of each specimen was initially measured 11–13 months after collection. Three hundred and twenty-four specimens were selected from the normal, reversed, and transition zones. In this article, the transition interval is considered to be repre-

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

Table 1

Mean NRM directions of selected low intensity specimens

Polarity	N		t	R	K	α_{95}	Dec.	Inc.
Normal	12	*a*	0	9·69	4·76	22·3	−4·0	22·4
		b	6	8·35	3·0	30·3	3·0	19·4
Reversed	6	*a*	0	5·43	8·92	23·7	158·4	−11·2
		b	4	5·39	8·21	24·9	156·4	−12·2
Transition	59	*a*	0	36·55	2·58	14·7	−43·4	18·9
		b	6	36·49	2·57	14·7	−45·0	3·2

a = Initial measurement; *b* = measured '*t*' months after the initial measurements. For explanation of symbols see caption to Table 3.

Table 2

Mean NRM intensities ($\times 10^{-4}$ emu cm^{-3}) *for polarity zones of long core*

Polarity	A	B	C	CT
N_2	0·127	0·138	0·147	0·117
R_2	0·092	0·092	0·093	0·092
R_3	0·122	0·128	0·132	0·124
R_2, R_3	0·117	0·120	0·124	0·117
TR	—	0·028	—	—

TR = Transition interval, see text for detailed explanation; A, B, C and CT are same as in Table 3.

sented by those samples whose palaeomagnetic properties are neither normal nor reversed and are located at a polarity boundary. It may or may not represent the true geomagnetic transition interval.

The specimens used in the storage test represent a number of diverse lithologies (Fig. 4). Many in the group yielded relatively weak NRM intensities whose initial directions showed large deviations from the mean directions. The NRM of these specimens were repeatedly measured over a period of 4–7 months. Between these repeat measurements all specimens were stored in the laboratory oriented such that their initial NRM vectors were approximately 90° away from the Earth's field. Their NRM directions were measured before and after storage for periods ranging from 3 to 30 days in a region where the residual field was less than about 200 gammas.

All normally and reversely magnetized specimens whose initial NRM directions did not deviate significantly from the mean directions showed subsequent changes of less than 2° during storage tests. The low intensity specimens, however, were found to have directions which changed significantly during the storage tests. No correlation was observed between the latter type of sample and either its lithology or magnetic polarity. The overall increase in dispersions for normal and reversed zones were statistically insignificant (Table 1), according to the comparison of directional dispersions (Watson 1956). Less than 5 per cent of the specimens from the normal and reversed zones were unstable. A few of the typical changes in direction by these unstable specimens are shown in Figs 5 and 6.

Many specimens from the transition intervals had intensities of less than $0{\cdot}2 \times 10^{-5}$ emu cm^{-3}. Fifty-nine specimens, which represent about 50 per cent of the weakly magnetized specimens, changed their directions during the storage test. Their directions were originally randomly distributed and remained so after the test (Table 1).

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

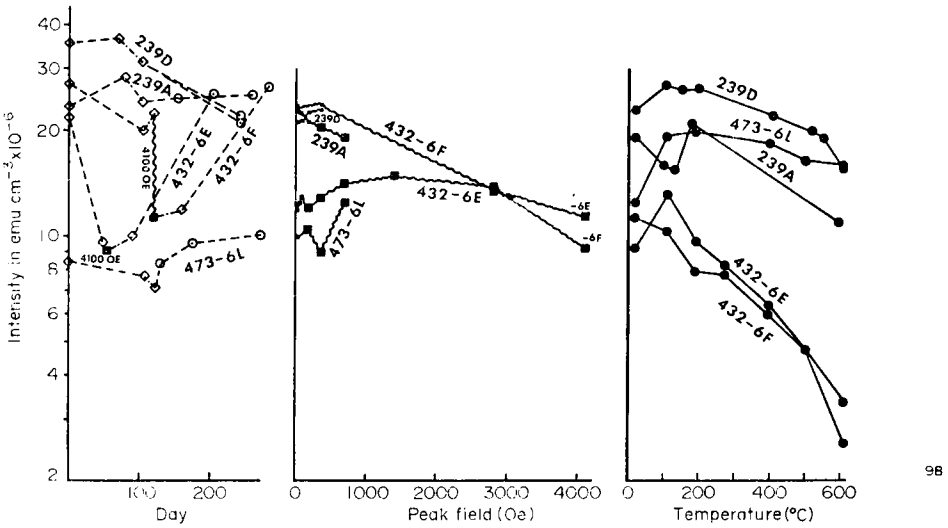


FIG. 5. Intensity changes of selected low intensity specimens. The dashed line is for a storage test in the Earth's field; dot and dash is storage test in zero field; wavy line is AC demagnetization; solid line is thermal demagnetization.

The average NRM intensity ratio of normal to reversed zones for subsurface specimens is approximately unity (Table 2). However, the average NRM intensity of specimens from transition intervals is about 0·25 of that of the reversed or normal zones.

After the test period of 4–6 months, the average intensities of specimens of the normal and reversed zones decreased by about 4·1 and 2·2 per cent, respectively, while the intensities of specimens belonging to the transition zones increased by 10·2 per cent.

(b) *AF demagnetization*

To further test the stability of the NRM, AF demagnetization was performed on stable and unstable specimens. The demagnetization apparatus used was the same as that used in the surface rock study by Helsley (1969). The progressive AF demagnetization had little effect on the stable NRM direction. An applied field of 350 Oe. was selected for 83 weakly magnetized specimens used in the storage test. Only 13 of the specimens showed direction changes of more than 10° and generally less than 20° in either declination or inclination. This change in magnetization direction usually resulted in a decrease in scatter. The other 70 specimens showed little change in their magnetization directions. Figs 5 and 6 illustrate the effects of AC demagnetization on five specimens having weak magnetization intensities.

(c) *Thermal demagnetization*

Twenty-four pilot specimens were heated progressively to 650 °C at approximately 100 °C intervals and cooled in a field-free space (± 20 gamma). Specimens from the transition intervals changed their directions after every increment up to about 500 °C. Specimens belonging to the normal and reversed zones did not change their directions above 400 °C. The selected specimens from the stable N_2, R_3, and R_2 zones showed changes in their mean magnetization directions of 5·3°, 3·5° and 2·6°, respectively, after thermal demagnetization at 510 °C to 610 °C (Table 3). Surface specimens from R_3 and R_2 zones showed changes about 26·4° and 13·2°, respectively, in their mean directions, after thermal demagnetization (Table 3).

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

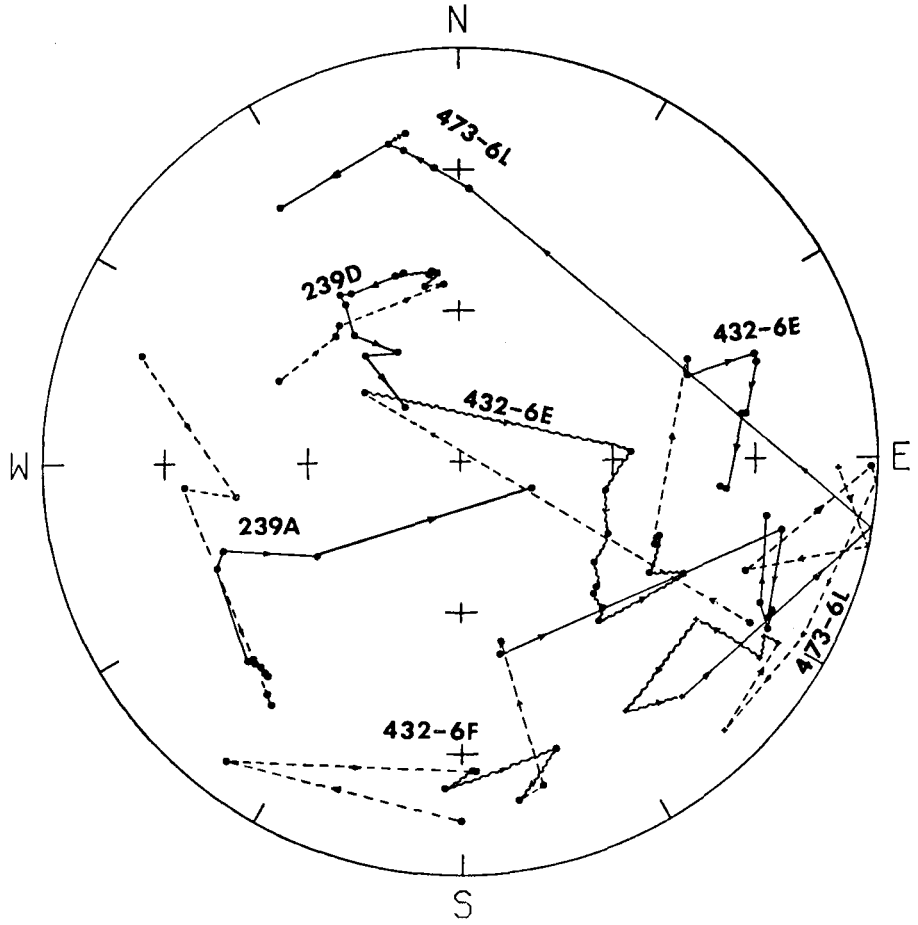


FIG. 6. Stability test. Direction changes of selected low intensity specimens. Line symbols are same as in Fig. 5.
⊙, lower; +, upper.

One hundred and fifty-three specimens taken from the four long segments of the core (segments 133, 135, 211 and 457) showed mean direction changes of about 6·8°, 11·7°, 3°, 0·9°, respectively. Specimens from segment 135 showed slightly increased scatter, while in the others, the scatter decreased slightly after thermal demagnetization. The results of thermal demagnetization suggest that the subsurface samples may have acquired some unstable secondary components during the approximately one-year period between their collection and their measurement. These could not be removed by AC demagnetization, but were easily removed by thermal demagnetization.

Estimates of concentration parameters for different lithologic units

All segments of the core longer than 10 cm, with the exception of the transition intervals were magascopically classified into four groups based on their grain size. These are: (1) siltstone, (2) fine sandstone, (3) coarse sandstone, and (4) conglomerate. Estimates of concentration parameters, K (Fisher 1953) for all core segments are plotted in Figs 7 and 8 along with the NRM intensities of these core segments.

For the long segment 457 consisting of 56 fine sandstone specimens we have chosen to randomly select 5, 10, 15, ... etc. specimens (Fig. 9). This selection was repeated a number of times and the estimates of the concentration parameter, K and

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

Table 3

Remanent magnetizations of normal (N_2) and reversed (R_2 and R_3) zones and the corresponding zones from surface sampling of the Moenkopi Formation

Polarity		Dec.	Inc.	N	R	K	α_{95}	δ	P. Long. (E)	P. Lat. (N)	dp	dm
	A	159·5	−11·6	351(97)[3]	335·900	23·2	1·6	16·8	−74·1	−52·4	0·8	1·6
R_3	B	158·3	−11·2	236(28)	226·950	26·0	1·8	15·9	−72·5	−51·7	0·9	1·9
	C	159·3	−10·6	50(28)	48·594	34·9	3·5	13·6	−74·2	−51·9	1·8	3·5
	CT	161·2	−13·6	48(27)	46·803	39·3	3·3	12·8	−76·0	−54·1	1·7	3·4
	A	351·9	24·0	220(73)	202·254	12·3	2·8	23·0	88·7	63·1	1·6	3·0
N_2	B	351·7	20·8	135(18)	127·050	16·9	3·1	19·7	88·2	61·3	1·7	3·2
	C	351·7	21·5	25(18)	24·056	25·4	5·9	15·8	88·4	61·7	3·3	6·2
	CT	347·5	18·0	24(17)	23·166	27·6	5·7	15·7	95·3	58·6	3·1	6·0
	A	165·0	−18·7	65(2)	64·201	80·1	2·0	9·0	−80·1	−58·1	1·1	2·1
R_2	B	165·0	−18·7	65(2)	64·201	80·1	2·0	9·0	−80·1	−58·1	1·1	2·1
	C	166·4	−17·5	14(2)	13·866	96·8	4·1	7·9	−83·0	−58·1	2·2	4·2
	CT	164·4	−19·2	14(2)	13·914	150·4	3·3	6·4	−78·9	−58·2	1·8	3·4
	A	160·4	−12·7	416(99)	399·451	25·1	1·4	16·2	−75·0	−53·4	0·7	1·4
R_2, R_3	B	159·7	−12·8	301(30)	290·402	28·3	1·6	15·2	−74·0	−52·1	0·8	1·6
	C	160·8	−12·1	64(30)	62·301	37·1	3·0	13·2	−75·9	−53·3	1·5	3·0
	CT	162·0	−14·9	62(29)	60·649	45·2	2·7	12·0	−76·6	−55·0	1·4	2·8
R_3^{S} [1]		166·4	5·2	34	31·39	12·7	7·2	22·4	−88·8	−46·9	3·6	7·2
N_2^{S} [1]		343·3	27·8	13	12·390	19·7	9·6	17·6	107·3	62·0	5·7	10·5
R_2^{S} [1]		165·8	6·9	45	38·960	7·3	8·5	29·7	−88·2	−45·9	4·3	8·5
R_3^{SST} [2]		165·7	−21·2	12	11·818	60·3	5·6	10·0	−80·4	−59·7	3·1	6·0
R_2^{SST} [2]		162·6	−5·9	38	36·447	23·8	4·9	16·4	−80·6	−51·0	2·5	4·9

Dec. = declination, positive east of north; Inc. = inclination, positive downward; N = number of specimens; R = vector sum of N unit vectors; K = concentration parameter estimate (Fisher 1953); α_{95} = semiangle of 95 per cent confidence cone (Fisher 1953); δ = angular standard deviation (Wilson 1959); P. Long. (E) and P. Lat. (N) = longitude and latitude of mean pole position, calculated using mean direction; dp and dm = length of semiaxes of 95 per cent confidence oval; see text for A, B, C, CT.

Note: 1. NRM data from surface rocks (Helsley 1969).

2. After 500 C thermal demagnetization (scattered data removed) of surface rocks (Helsley 1969). Values given in this table are corrected for errors in Helsley's (1969) table.

3. Number of independently oriented core segments are in parenthesis.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

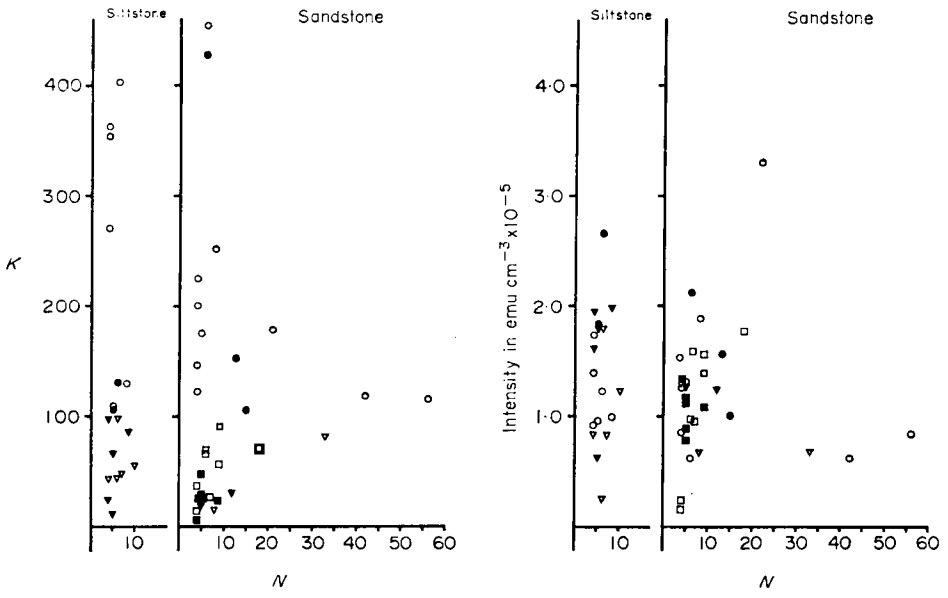


FIG. 7. Estimates of Fisher's (1953) concentration parameter, K, and average NRM intensities for siltstone (○, ▽), fine sandstone (○, ▽) and coarse sandstone (□) segments longer than 10 cm. Circles are for segments with K values greater than 100. N is number of specimens for each core segment. Solid symbols are normal polarity.

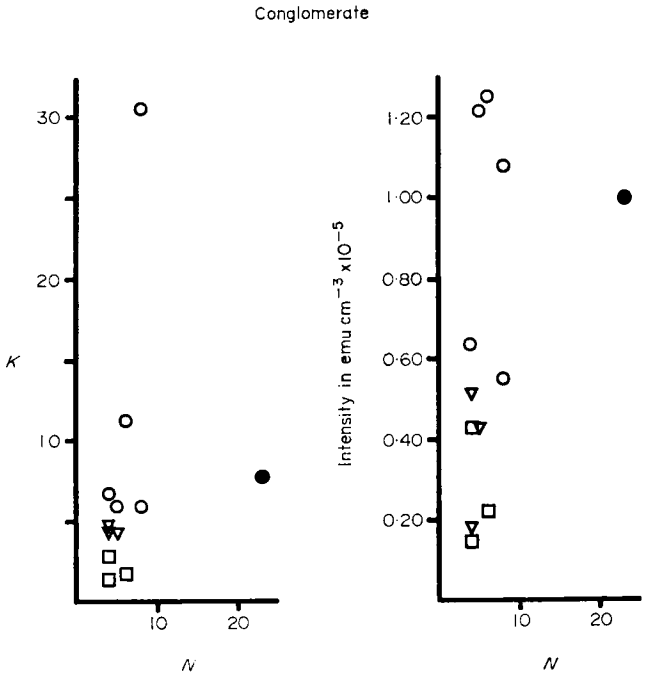


FIG. 8. The estimates of Fisher's (1953) concentration parameter and average NRM intensities for 12 conglomeratic segments longer than 10 cm. Open square is for the segment whose within-segment NRM directions are tested random at the 95 per cent confidence level; circles are for K greater than 5. Solid symbol is normal polarity.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

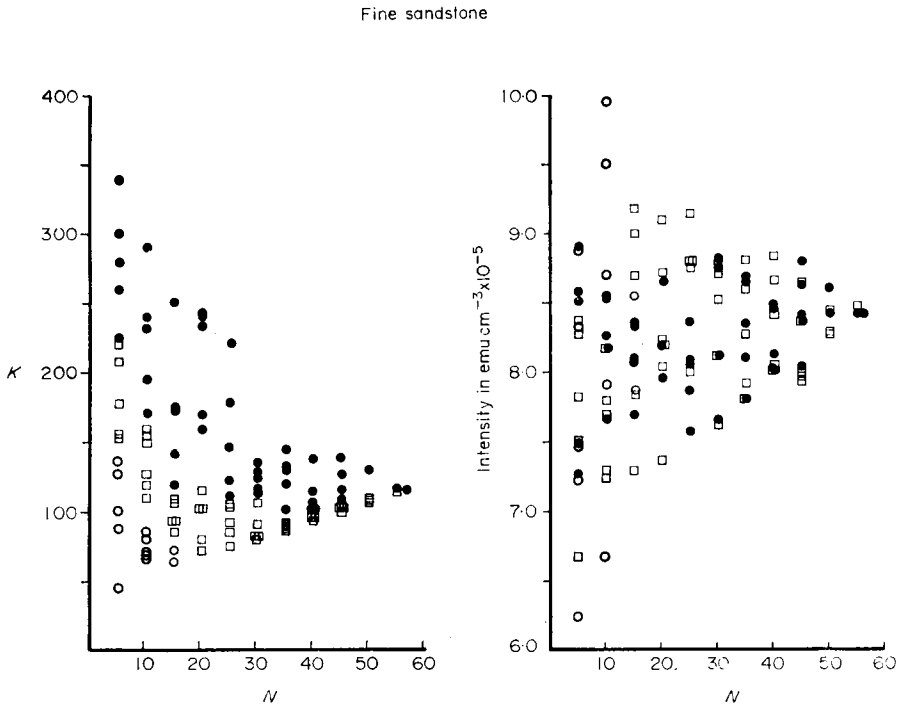


FIG. 9. Estimates of Fisher's (1953) concentration parameter, K, and average NRM intensities for randomly selected sets of specimens from segment 457. N is size of the subset. Sets for $N = 5$, 10 and 15 are arbitrarily classified into three groups (solid circle, open square and open circle) according to relative K values; for $N \geqslant 20$, they are classified into two groups. See text for detailed discussions.

average NRM intensities were calculated for each selected set and are compared in Fig. 9. The specimens sets having large K values (solid circles in Fig. 9) do not exhibit extremely low or high average intensities. In contrast to the close relationship found by Irving & Runcorn (1957), little correlation was observed between the concentration parameters and NRM intensities with this core segment.

In Fig. 9 it is worthwhile to note the large variations in K values for the smaller sized specimen subsets. This indicates that in palaeomagnetic studies there can be uncertainties in estimates of the concentration parameters for small numbers of samples. This uncertainty may be caused by the statistical dependence existing between palaeogmagnetic samples, the nature of which is generally not known. For clarity the confidence limits for the concentration parameter (Cox 1969) are not given in Figs 7, 8 and 9, but can be calculated since the number of samples and the K values are both indicated.

All segments of siltstone, except four ($K > 270$), have low K values which are similar to those of the coarse sandstones. The low K values for very fine grained siltstone may be caused by a superparamagnetic component. None of the fine sandstone units have low K values and the fine sandstone specimens generally give the largest K values. Twelve conglomerate segments which did not belong to transition intervals gave the smallest K values. A similar dependence of the concentration parameters on grain size was found by Irving & Runcorn (1957) and was discussed more completely by Runcorn (1967).

Within the conglomerate segments only three were found to be random at the 5 per cent significance level. Of the 12 conglomerate segments, eight had stable

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

NRM, and the mean of these is not significantly different from that from the finer grained portions of the core.

Discussion

(a) *Statistical analyses*

There are problems encountered in statistical studies of sedimentary rock properties, because sedimentary rock specimens often cannot be considered to have been randomly drawn from the same population. In this study, lithologic differences, variable deposition rates, different rates of sample recovery, lengths of each independently oriented core segment, and depth of core segment can all cause possible statistical bias. This can result in errors in the calculated mean directions and underestimates of their cones of confidence. Most of these factors are unknown variables and we know of no statistical method which can satisfactorily allow for them.

The longer the core segment the smaller will be the error of orientation. A number of short core segments representing the same interval as a single long segment will tend to average out the orientation errors. However, the NRM directions obtained from these segments will have a significantly larger dispersion, if the specimens are treated with equal weight.

Four sets of specimens for each polarity zone were selected, with the exception of conglomerates, in the following four ways:

(A) Specimens from all core segments longer than 2 cm.

(B) Specimens from segments longer than 10 cm.

(C) One specimen from each segment longer than 10 cm plus an additional specimen for each further 12 cm of length.

(CT) Thermally demagnetized specimens, most of which belonged to the set C.

In Table 3, mean directions, statistical analyses and pole positions are listed for the above four sets of selections with the demagnetized results of corresponding surface rocks (Helsley 1969).

Remarkably little difference is found in the mean directions among the three NRM sets of (A), (B) and (C). However, there is a large difference in the cones of confidence and the angular standard deviations (Wilson 1959) as would be expected.

The long core did not penetrate the entire R_2 zone. As a result, only two independently oriented core segments which total about 1·6 m were recovered from this zone. This is a small portion (about 7·6 per cent) of the entire R_2 zone which from surface samples (Helsley 1969) is considered to be about 21 m thick. Therefore, from a statistical point of view, a comparison of the surface and subsurface magnetic data for this zone is not very significant.

(b) *Correlation with surface data*

The top of the hole at the drilling site was about 13 m below the N_3/R_3 boundary (Fig. 2 of Helsley 1969). This reversal boundary is exposed about 200 m north-west of the drilling site.

The stratigraphic plot of the subsurface NRM and that of corresponding surface data (Helsley 1969) are illustrated on Fig. 10. The differences in thickness between subsurface and surface magnetozones are due to the local wedging of the formation. Of the 1117 specimens, the magnetic data for 63 have been omitted from Fig. 10 because the specimens were of insufficient size. These 63 specimens, as well as those belonging to intervals of polarity transition and conglomeratic specimens, have all been excluded from Fig. 11. This figure is an equal area stereographic plot of the NRM data.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

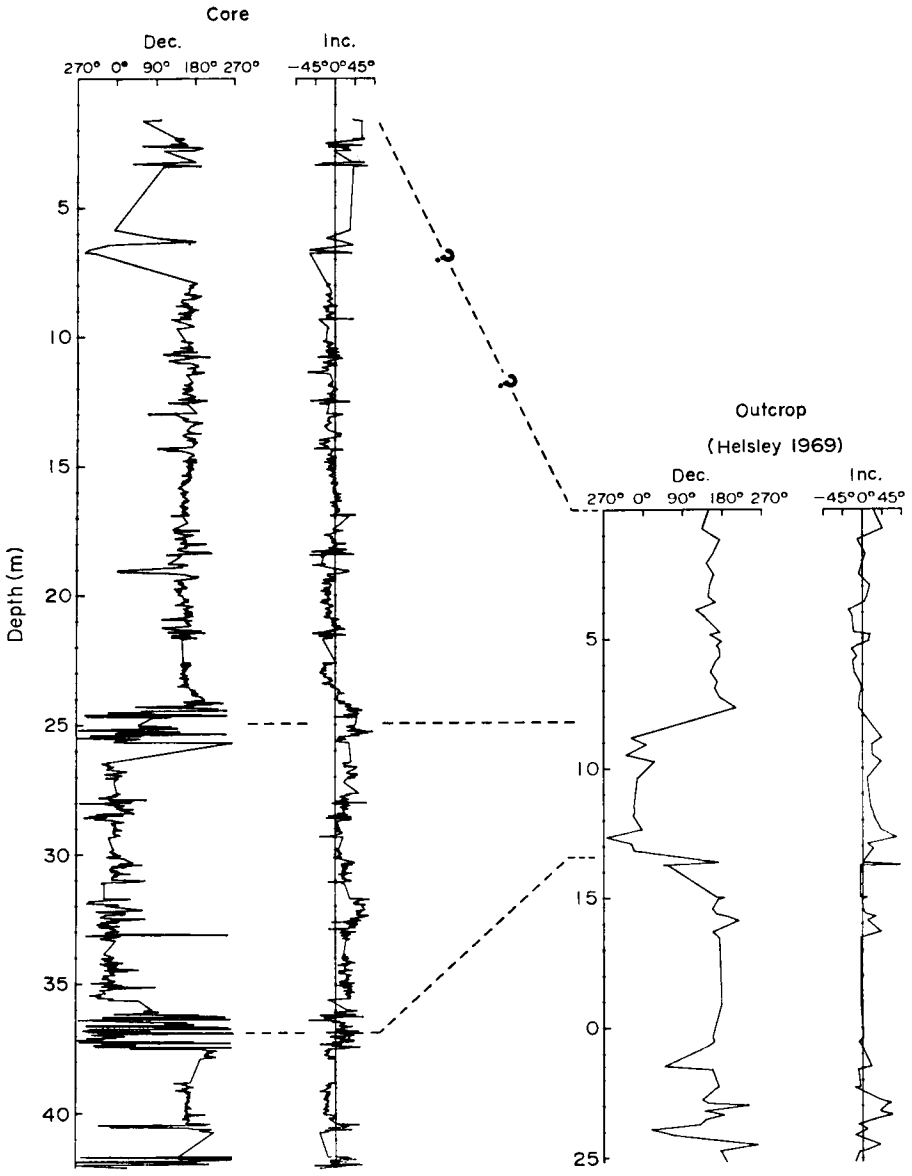


FIG. 10. Stratigraphic plot of NRM declinations and inclinations of all specimens with the exception of 63 of very small size. These data are compared to corresponding surface NRM data of Helsley (1969).

In the surface outcrop study (Helsley 1969), the degree of scatter of NRM from normal polarity zones was markedly smaller than that for reversed zones. This feature was not evident in measurements on the subsurface core. Little difference in the dispersions calculated for segments longer than 10 cm representing both normal and reversed zones was observed.

(c) *Large inclination variations*

The stratigraphic plots of NRM declinations and inclinations of four unbroken long segments of the core (Nos 133, 135, 211 and 457) are compared in Fig. 12. The

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

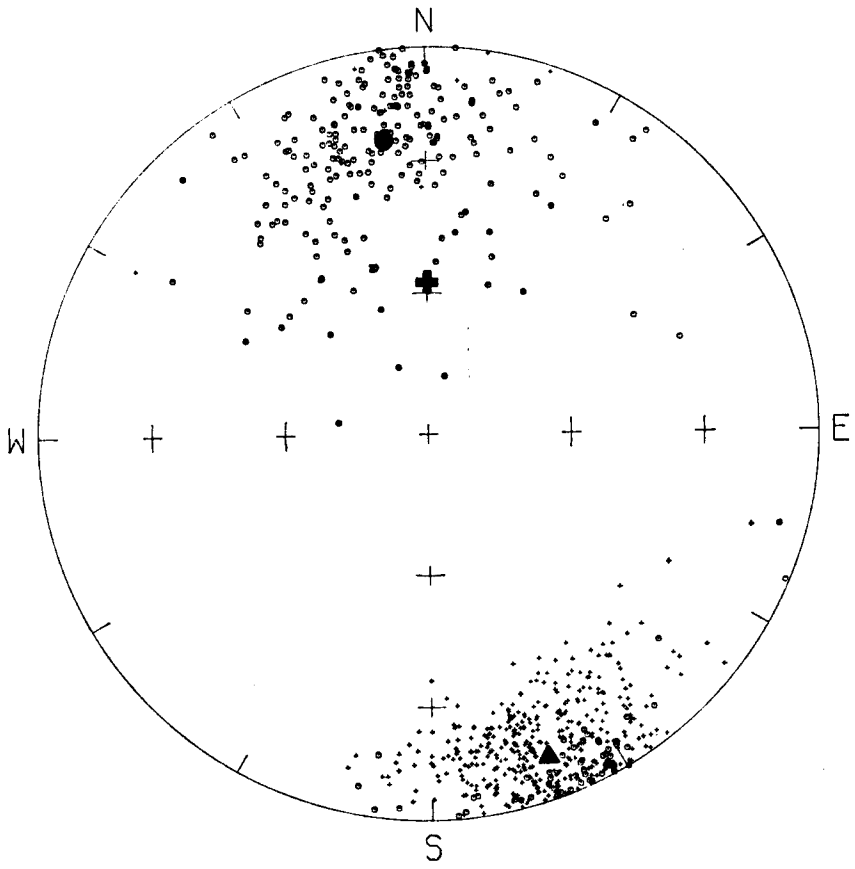


FIG. 11. Winkie NRM. Equal area plots of the NRM directions for the normal (N_2) and reversed (R_2 and R_3) zones. Conglomerates and specimens with uncertain orientation and/or very small size are excluded. 'Present' is present geocentric axial dipole field direction at the drilling site.
⊙, lower; +, upper; ▲●, mean; +, present.

maximum difference in the mean declinations of the four segments is about 12° for NRM and is also 12° after thermal demagnetization. The maximum difference in mean inclinations is about 30° for NRM and 22° after thermal demagnetization. The 12° deviations in average declination suggest again that the orientation error introduced by the ellipticity of the bedding plane method is less than ±6°.

The range of 22° in the mean inclination values for the four long core segments could be caused by:

(1) Gradual change of drilling angle with depth.

(2) Local bedding error (King 1955).

(3) Actual field changes.

Segments 133, 135 and 211 of the core belong to the R_3 zone and 457 belongs to the upper part of the R_2 zone. Segments 133 and 457 are separated by 28 m (Fig. 4). Segments 211 and 457 have steeper average inclinations. Because the average magnetic inclination of 211 is steeper, rather than shallower, than that of 457, which lies 18 m *below* it, a postulated gradual change of drilling angle cannot be the only cause of the large inclination variations. To investigate this further, the angles

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

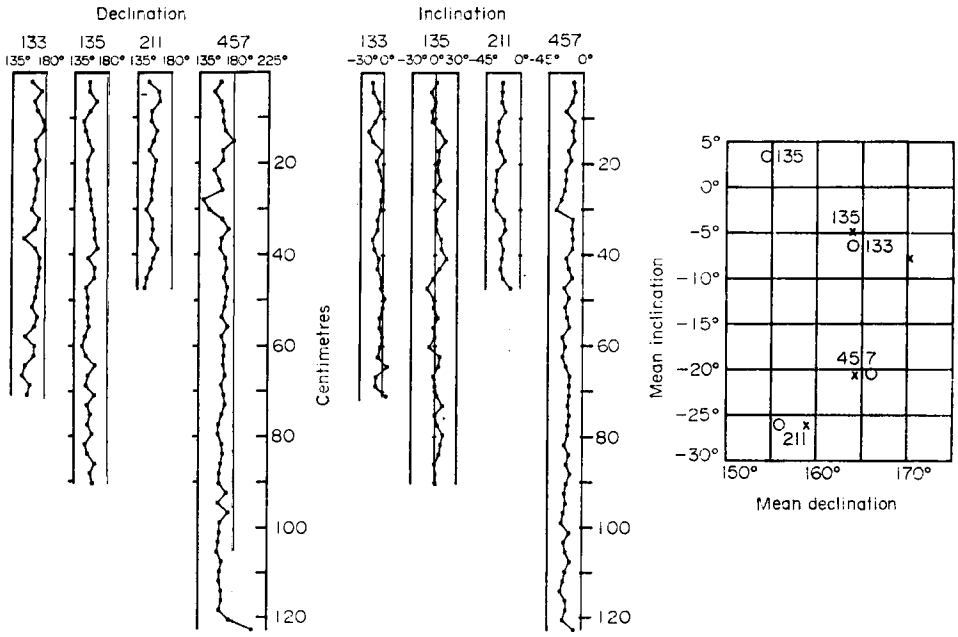


FIG. 12. Stratigraphic plots of NRM declinations and inclinations of core segments 133, 135, 211 and 457. Open circle is NRM, X is after thermal demagnetization.

between bedding planes and the sides of the specimens were measured for a total of 27 specimens which exhibited very clear bedding planes. Three measurements were made for each of the 27 specimens. Fifteen of the specimens came from the upper part of the hole and gave an angle of $53{\cdot}3° \pm 1{\cdot}6°$ and 12 specimens came from the lower part of the hole and gave an angle of $59{\cdot}5° \pm 3{\cdot}6°$. This indicates that the maximum change in either the inclination of the hole or the dip of the formation with depth is a few degrees and thus this effect is insignificant relative to the 22° variation observed in the magnetic inclinations. The angle 58° was applied to all specimens for core orientation correction in NRM computation after the remanent magnetization directions were measured with respect to the specimen's co-ordinate system. Since the possible effects of even the small deviations of the resultant NRM due to the variation of the actual drill hole inclination from the assumed inclination of 58° may be a contribution to the 22° variation they are discussed below in some detail, as some of the results are not immediately obvious.

The simple rotational transformation matrix computation yields the expression:

$$I = \frac{\pi}{2} - \arccos(\cos\gamma \sin\theta - \cos\alpha \cos\theta) \quad \text{(in radian)} \tag{1}$$

$$\{0 \leqslant \arccos(\cos\gamma \sin\theta - \cos\alpha \cos\theta) \leqslant \pi\}$$

where $\cos\gamma$, $\cos\alpha$ are the direction cosines of NRM along the samples vertical axis (Z) and in the direction of the orientation mark (X), respectively, θ is the drilling angle, taken as being positive when measured downwards from the horizontal, I is the corrected inclination.

Differentiating (1) gives:

$$\frac{\partial I}{\partial \theta} = \frac{\cos\alpha \sin\theta + \cos\gamma \cos\theta}{\cos I}. \tag{2}$$

The denominator of (2) is always positive because $0 \leqslant |I| \leqslant \pi/2$. The numerator

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

can give four possible regions with $0 \leqslant |\theta| \leqslant \pi/2$, namely:

$$\cos\gamma > 0, \cos\alpha > 0\text{: (a)}\frac{\partial I}{\partial\theta} > 0, \quad \text{if} \quad I > 0\text{; (b)}\frac{\partial|I|}{\partial\theta} < 0, \quad \text{if} \quad I < 0$$

$$\cos\gamma < 0, \cos\alpha < 0\text{: (c)}\frac{\partial|I|}{\partial\theta} > 0, \quad \text{if} \quad I < 0\text{; (d)}\frac{\partial I}{\partial\theta} < 0, \quad \text{if} \quad I > 0$$

$$\left.\begin{array}{l}\cos\gamma > 0, \cos\alpha < 0\\ \cos\gamma < 0, \cos\alpha > 0\end{array}\right\} \qquad \frac{\partial I}{\partial\theta} \gtrless 0.$$

(Note: For the first two regions, the corrected inclination, I can be both positive and negative; for the last two regions sign of I will remain unchanged for any value of $0 \leqslant \theta \leqslant \pi/2$.)

The normal and reversed magnetization directions of this subsurface study belong to the first two regions. Thus, if the angle of 55·3° rather than 58° represents the actual drilling angle at the upper part of the hole, the result of over correction of the angle of the hole would give steeper NRM inclinations for the specimens from the upper part of the hole. This effect is exactly opposite to the measured average NRM variations which increase with depth of hole. However, such errors, introduced by a change of a few degrees in the relative drilling angle, are insignificant relative to the 22° variation observed in the average inclinations.

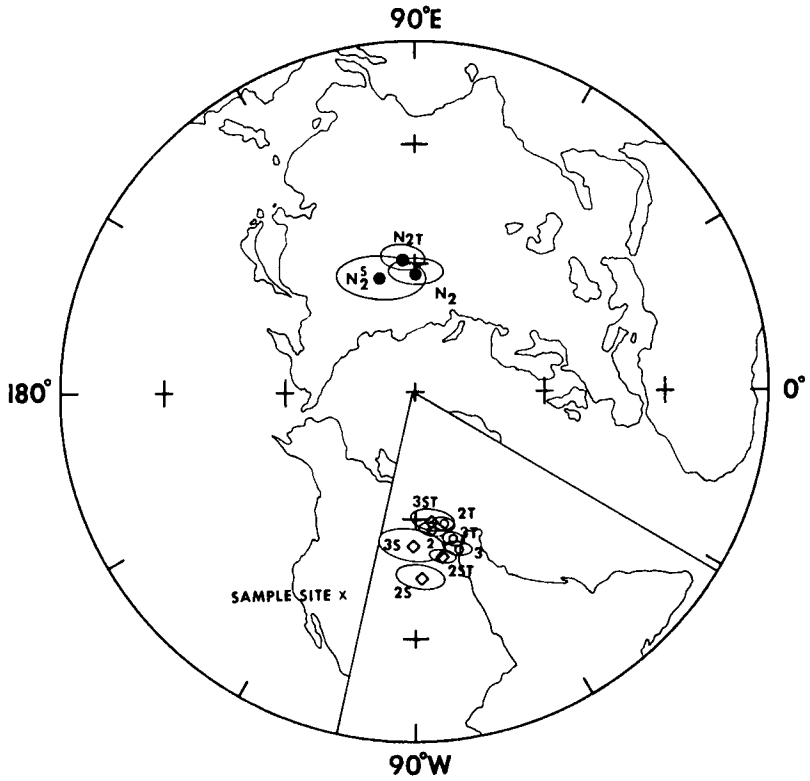


FIG. 13. Pole positions from magnetozones N_2, R_2 and R_3 derived from subsurface data compared with pole positions of corresponding surface samples. Pole positions N_2, N_{2T}, N_2^S (solid circles) represent N_2(C), N_2(CT) and N_2^S in Table 1; positions 3, 3T, 2 and 2T (open circles) represent R_3(C), R_3(CT), R_2(C) and R_2(CT); positions 3S, 2S and 2ST (open squares) represent R_3^S, R_2^S and R_3^{SST}, respectively.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

The magnetic inclination errors resulting from sedimentologic parameters have been discussed by King (1955) whose experimental studies indicated that some of the inclination error associated with sedimentary rocks was directly related to the initial bedding angle. The errors arising from changes in the relative angles of the depositional surfaces between segments is probably insignificant since the depositional surface probably remained constant over the 28 m of section involved. Creer (1970) has stated that inclination errors can exist in red beds. However, the presence of the inclination error in the Moenkopi red beds has yet to be demonstrated. We therefore feel that changes in the palaeogeomagnetic field directions are the primary cause of the observed differences in inclination.

(d) *Pole positions*

The pole positions calculated from the mean directions before and after thermal demagnetization for each polarity zone are given in Table 3 and are plotted in Fig. 13. The pole positions of corresponding surface data are also shown in Table 3. The NRM poles for the surface and subsurface normal zone (N_2) are not significantly different (Watson's (1956) equivalence test gives $F_{2,\,72} = 2{\cdot}8$). However, the NRM poles for the R_2 and R_3 zones are significantly different. As discussed earlier in the comparison of the mean NRM directions, the poles for R_2 and R_3 cannot be compared for the surface and the subsurface data. It is interesting to note that the surface poles for R_2 and R_3, obtained after thermal demagnetization, lie between the corresponding surface NRM poles and the subsurface NRM poles. This suggests that the demagnetized surface results still have more secondary magnetization than do the NRM results of subsurface samples which is the basis for our conclusion that most 'secondary' magnetization in the Moenkopi Formation is acquired as a result of surface effects (e.g. weathering).

Conclusions

The NRM directions of the Moenkopi red bed long core cluster more tightly than those of samples from surface outcrops. Although the inclination measurements from the surface samples did not provide distinctive polarity change characteristics, the analyses of inclinations from the long core specimens did.

The orientation error introduced by the ellipticity of bedding section is not considered significant and the dispersion due to this error is smaller than that caused by surface effects. Randomly selected subsurface core samples from the N_2, R_3 and R_2 magnetic zones showed that the changes in the mean directions after thermal demagnetization were 5·3°, 3·5° and 2·6°, respectively. The surface samples showed changes of 26·4° and 13·2° for the R_3 and R_2 zones after thermal demagnetization. The NRM direction of the N_2 normal zone is statistically equivalent to that of the equivalent surface samples. Subsurface core segments from the R_2 zone represent a small portion of the entire R_2 zone, thus comparison of the mean surface and subsurface directions of the R_2 zone is not very meaningful.

The estimates of the within-segment concentration parameters become smaller as the grain size increases. The dependence of concentration parameter on grain size may indicate dependence of NRM on the rate of deposition, differences in magnetic and other depositional environments and/or diagenesis. NRM intensity generally becomes weaker with increase in grain size. The short segments of the core, consisting of very fine, brown sandstone and siltstone, from the N_2 normal zone, showed very high average intensity. The NRM intensity ratio of normal : reversed : transition zones for the core segments longer than 10 cm are 1·2 : 1·0 : 0·2.

The palaeomagnetic results of this study indicate that subsurface samples oriented by means of the elliptical bedding section can be used to derive reliable estimates of pole positions.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

Acknowledgments

The drilling was performed by C. E. Helsley, W. Gose and M. Steiner with assistance of P. J. Helsley; their help is gratefully acknowledged. Valuable assistance from M. Steiner and White Chaipayungpun in carrying out the magnetic measurements is sincerely appreciated. This work was supported by NSF grant GA 15999 to C. E. Helsley.

Institute for Geological Sciences,
The University of Texas at Dallas,
Box 30365, *Dallas, Texas* 75230.

References

Briden, J. C., 1965. Ancient secondary magnetizations in rocks, *J. geophys. Res.*, **70,** 5205–5221.

Briden, J. C., 1967. Secondary magnetization of some Paleozoic rocks from Tasmania, *Pap. Proc. R. Soc. Tasm.*, **101,** 43–48.

Briden, J. C. & Ward, M. A., 1966/I. Analysis of magnetic inclination in borecores, *Pure appl. Geophysics*, **63,** 133–152.

Chamalaun, F. H., 1964. Origin of the secondary magnetization of the Old Red Sandstones of the Anglo-Welsh Cuvette, *J. geophys. Res.*, **69,** 4327–4337.

Cox, A., 1969. Research note, confidence limits for the precisions parameter *K*, *Geophys. J. R. astr. Soc.*, **17,** 545–549.

Creer, K. M., 1970. A review of paleomagnetism, *Earth Sci. Rev.*, **6,** 369–466.

Farrell, W. E. & May, B. T., 1969. Paleomagnetism of Permian redbeds from the Colorado plateau, *J. geophys. Res.*, **74,** 1495–1504.

Fisher, R., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, **A217,** 295–305.

Graham, K. W. T. & Hales, A. L., 1961. Preliminary palaeomagnetic measurements on Silurian sediments from South Africa, *Geophys. J. R. astr. Soc.*, **5,** 318–325.

Helsley, C. E., 1969. Magnetic reversal stratigraphy of the lower Triassic Moenkopi Formation of western Colorado, *Geol. Soc. Am. Bull.*, **80,** 2431–2450.

Helsley, C. E. & Steiner, M., 1974. Magnetic reversal stratigraphy and secular variation in the Lower Triassic Moenkopi Formation of western Colorado, *Geol. Soc. Am. Bull.*, in press.

Irving, E., 1964. *Paleomagnetism and its application to geological and geophysical problems*, John Wiley and Sons, Inc., New York.

Irving, E. & Opdyke, N. D., 1965. The paleomagnetism of the Bloomsburg red beds and its possible application to the tectonic history of the Appalachians, *Geophys. J. R. astr. Soc.*, **9,** 153–167.

Irving, E. & Runcorn, S. K., 1957. Palaeomagnetic investigations in Great Britain, II. Analysis of the palaeomagnetism of the Torridonian Sandstone Series of northwest Scotland. I, *Phil. Trans. R. Soc. Lond.*, **A250,** 83–99.

King, R. F., 1955. The remanent magnetism of artificially deposited sediments, *Mon. Not. R. astr. Soc. Geophys. Suppl.*, **7,** 115–134.

Ku, C. C., Sun, S., Soffel, H. & Scharon, L., 1967. Palaeomagnetism of the basement rocks, Wichita Mountains, Oklahoma, *J. geophys. Res.*, **72,** 731–737.

Picard, M. D., 1964. Paleomagnetic correlation of units within Chugwater (Triassic) Formation, west-central Wyoming, *Am. Assoc. Petroleum geol. Bull.*, **48,** 269–291.

Runcorn, S. K., 1967. Statistical discussion of magnetization of rock samples, *Methods in palaeomagnetism*, pp. 329–339, eds. D. W. Collinson, K. M. Creer & S. K. Runcorn.

Shoemaker, E. M., 1955. Geology of the Juanita Arch Quadrangle, Colorado, *U.S. Geol. Survey. Quad. Map* 81.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022

Shoemaker, E. M. & Newman, W. L., 1959. Moenkopi Formation (Triassic? and Triassic) in salt anticline region, Colorado and Utah, *Am. Assoc. Petroleum geol. Bull.*, **43**, 1835–1851.

Storetvedt, K. M., 1968. On remagnetization problems in palaeomagnetism, *Earth Planet. Sci. Lett.*, **4**, 107–112.

Watson, G. S., 1956. Analysis of dispersion on a sphere, *Mon. Not. R. astr. Soc., Geophys. Suppl.*, **7**, 153–159.

Wilson, R. L., 1959. Remanent magnetism of late Secondary and Early Tertiary British rocks, *Phil. Mag.*, **4**, 750–755.

Downloaded from https://academic.oup.com/gji/article/37/2/245/579263 by Curtin University Library user on 05 July 2022